Kun-Ying Lee

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Robot Learning | Humanoid Locomotion | Motion Generation

Education

Tsinghua University Sept. 2025–Jun. 2029 (expected)
Undergraduate, Yao Class GPA: 3.9/4.0
Computer Science (Artificial Intelligence Track)
Institute for Interdisciplinary Information Sciences (IIIS)

Selected Publications

OMG: Omni-Modal Motion Generation for Generalist Humanoid Control
Siqiao Huang*, **Kun-Ying Lee***, Dongming Qiao*, Guanqi He*, Zhenyu Wang, Yitang Li, Shaoting Zhu, and Hang Zhao.
arXiv:2606.10340, 2026. Paper | Project | Code
RSS 2026 workshops: ExWBC (**Oral**); RoboData (**Spotlight**).

Hiking in the Wild: A Scalable Perceptive Parkour Framework for Humanoids
Shaoting Zhu*, Ziwen Zhuang*, Mengjie Zhao, **Kun-Ying Lee**, and Hang Zhao.
Conference on Robot Learning (CoRL), 2026 (accepted). Paper | Project | Code

The Indefinite Integral of the *n*th Power of Inverse Trigonometric Functions
Kun-Ying Lee.
Mathmedia, Academia Sinica, 49(2):91–101, 2025. Article

* Equal contribution.

Research Experience

MARS Lab, Tsinghua University Nov. 2025–present
Research Assistant | Advisor: Prof. Hang Zhao

OMG-Posttrain Project Lead; ongoing

- Lead research on reinforcement learning post-training of a motion generator to enable a frozen motion tracker to execute highly dynamic skills beyond its baseline capabilities.
- Built residual PPO and noise PPO pipelines to improve closed-loop motion execution with a frozen tracker, integrating vectorized rollouts and independent execution-based evaluation.
- Investigating diffusion-based reinforcement learning to directly fine-tune the full motion-generation backbone using closed-loop execution feedback.

OMG: Omni-Modal Motion Generation Co-first author

- Built a video-to-robot data pipeline combining GVHMR motion reconstruction, humanoid retargeting, and VLM-based motion captioning to convert monocular dance videos into annotated robot-motion training data.
- Developed and pretrained a multimodal diffusion-transformer motion generator conditioned on language, audio, and human motion; conducted ablation studies of multimodal conditioning and attention mechanisms.
- Built an asynchronous closed-loop deployment system on Unitree G1, integrating ONNX/TensorRT inference, replanning from measured robot-state history, and temporal alignment between motion generation and control.

Hiking in the Wild: Perceptive Humanoid Parkour

- Validated sim-to-sim transfer of a depth-conditioned parkour policy to support deployment on Unitree G1.
- Resolved a sim-to-real perception gap caused by reflection-induced depth artifacts through invalid-depth masking, spatial inpainting, and Gaussian filtering, aligning real sensor inputs with training observations.
- Supported real-robot deployment and field testing; the system traversed stairs, slopes, and gaps, with running speeds up to 2.5 m/s and gap widths up to 50 cm.

Engineering Experience

TH-MOS Robot Soccer Team, Tsinghua University Sept. 2025–present
Technical Leader

Humanoid Locomotion and Soccer Skills

- Trained command-conditioned locomotion policies using adversarial motion priors (AMP), achieving stable real-robot velocity tracking over forward/backward velocities of $[-1.5, +2.0]$ m/s, lateral velocities of $[-1.5, +1.5]$ m/s, and yaw rates of $[-1.7, +1.7]$ rad/s on Booster K1.
- Developed shooting policies conditioned on target 3D coordinates within the goal mouth, evaluated from positions across a 14×9 m field, achieving targeting errors within 10 cm in 90% of physical trials (27/30). Exploring perception-guided curved shots to avoid obstacles.
- Trained a reference-free dribbling policy for Booster K1 using asymmetric PPO, jointly learning locomotion and head control from ball-velocity commands, proprioception, and camera-frame ball observations.
- Trained an AMP-based fall-recovery policy in MJLab that recovers from tested fallen poses to stable standing within 0.5 s, succeeding in all 30 real-robot trials.

THMOS-Simulation

- Developed a Unity–MuJoCo humanoid soccer platform supporting autonomous 3v3, 5v5, and 11v11 matches, integrating learned locomotion, dribbling, kicking, and recovery policies with particle-filter-based localization, YOLO-based object detection, and the HSL Game Controller.
- Investigating self-play multi-agent reinforcement learning for team decision-making, starting from MAPPO.

MOS-9 Open-Source Humanoid Robot Project

- Contributed to robot deployment, AMP-based locomotion training in Isaac Lab, motion-data retargeting, and MuJoCo sim-to-sim validation for MOS-9.
- Contributed to a carbon-fiber redesign that reduced robot mass from 16 kg to 11 kg.
- The project received the **Special Prize** at the National Additive Manufacturing Competition.

Open-Source Projects

SuperMimic on mjlab: Model-Based Humanoid Motion Tracking Paper | Code

- Reproduced SuperMimic's differentiable-world-model approach to humanoid motion tracking on native MuJoCo-Warp through mjlab, achieving 100% full-clip completion across 40 LAFAN1 motions within 5 hours of training on a single RTX 5090.
- Demonstrated few-shot adaptation of the pretrained tracking policy to highly dynamic MotionCode motions.

ScaleBFM-K1: Large-Scale Humanoid Motion Tracking Code

- Adapted BFM-style motion tracking to the 22-DoF Booster K1, achieving over 95% full-clip completion on the BONES-SEED test set after 5 hours of training on one node with 8 NVIDIA A10 GPUs.
- Deployed and verified the policy on the real robot.

Selected Honors & Awards

Ministry of Education Scholarship for Taiwan Students (**First Prize, Top 2 Schoolwide**) Oct. 2026
5th Place, RoboCup 2026 Incheon (TH-MOS team) Jul. 2026
2nd Place, RoboCup Asia-Pacific Tianjin Invitational Tournament (550W team) May. 2026
3rd Place, RoboCup China Open (TH-MOS team) May. 2026
Leading Talent (Lingjun) Award, Tsinghua University Oct. 2025

Technical Skills

Programming and Learning: Python, PyTorch
Simulation and Deployment: MuJoCo, Isaac Lab, ONNX Runtime, TensorRT